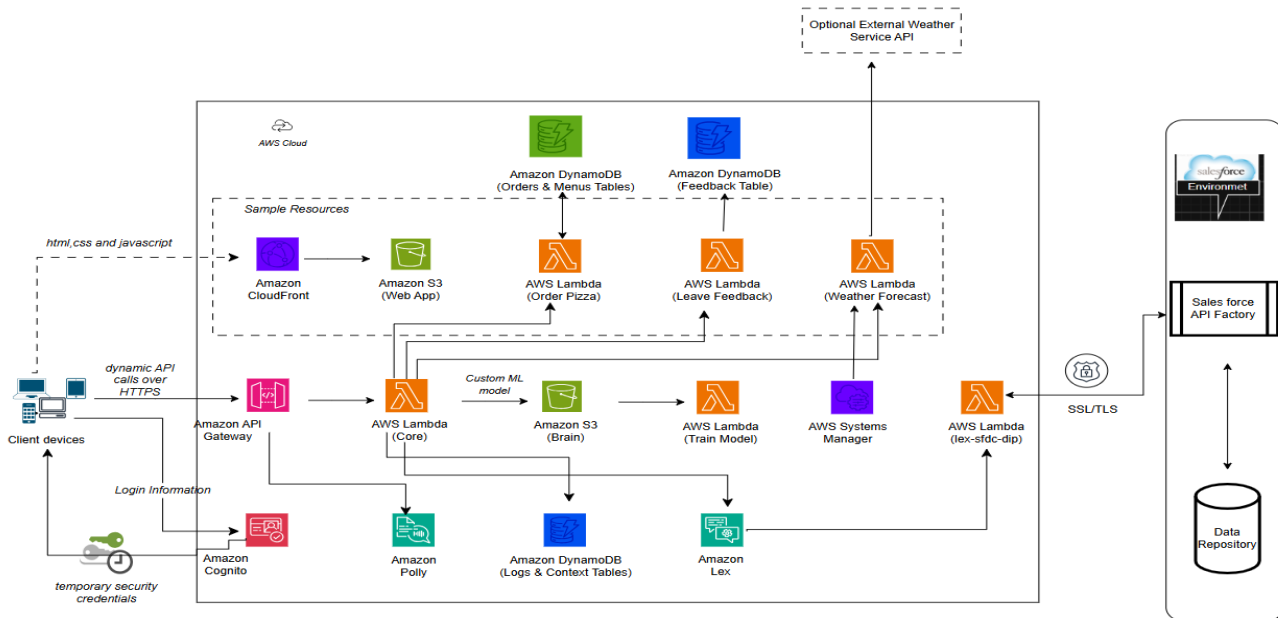




SYSTEM INVOLVED

- 1) **Customer:** Customer Uses a laptop (or) phone (or) tablet to open the app or website and type the order.
- 2) **Website (CloudFront + S3):** Cloud front is the website customer place the order
- 3) **Cognito:** Checks the customer and allow them a temporary security pass so that system authorise and understand the customer and without asking them to log in again.
- 4) **Front Desk:** Every request from the customer's device — whether it's "order a pizza" or "what's the weather" — comes through this one door first.
- 5) **Core Lambda:** It receives the request from the front desk and figures out what needs to happen next and which specialist to call.
- 6) **Amazon Lex:** when a customer says "I'd like a large pepperoni" instead of filling out a form, Lex works out what they mean (Understands language)
- 7) **Amazon Polly:** Turns the assistant's text replies into spoken audio, so the conversation can feel like talking to a real person.
- 8) **Lambdas (Order Pizza / Leave Feedback / Weather Forecast):** Focused workers, each responsible for exactly one job — placing an order, logging feedback, or fetching the weather.
- 9) **DynamoDB tables:** Where orders, menus, feedback, and conversation history are stored — quickly searchable and always up to date.
- 10) **(S3 "Brain" bucket + Train Model Lambda + Systems Manager):** A background process that keeps improving the assistant's understanding over time, based on past conversations.
- 11) **lex-sfdc-dip Lambda:** Securely sends relevant data like new orders or customer details into Salesforce, so it isn't stuck in AWS alone.
- 12) **Salesforce Environment:** CRM system, where sales and support team's work. It receives data through its own API Factory and stores it in its Data Repository.

Stories – STEP BY STEP**1. Customer opens the app:**

- ✓ He/she visit the pizza-ordering website or app on their phone or laptop.
- ✓ The page itself is delivered quickly through CloudFront (just fetching the website files stored in an S3 bucket).

2. Customer logs in:

- ✓ Amazon Cognito checks their login information and gives their device a temporary security pass.
- ✓ This pass proves who they are for the rest of the session, without needing to log in again for every single action.

3. Customer makes a request:

- ✓ The customer types something like "order me a large pepperoni pizza," "the delivery was late, here's my feedback," (or) "will it rain tonight?"
- ✓ This request travels over a secure connection (HTTPS) to Amazon API Gateway — the single front door for everything the app needs to do.

4. Core Lambda takes charge:

- ✓ API Gateway hands the request to the Core Lambda function, which acts like a receptionist deciding which department the customer needs.
- ✓ It also consults Amazon Lex to properly understand free-form text, and a custom machine-learning model (stored in the S3 "Brain" bucket) to make decisions, such as recommending a pizza based on past orders.

5. Right specialist is called:

- ✓ Based on what the customer wants, one of three focused Lambda functions takes over — Order Pizza, Leave Feedback, or Weather Forecast.
- ✓ Each does exactly one job and nothing else, which keeps the system simple to maintain and test.

6. Result is saved (or fetched):

- ✓ If it's an order, it's written into Orders & Menus DynamoDB table. If it's feedback, it goes into Feedback table.
- ✓ If it's a weather question, the Weather Forecast Lambda reaches out to an external weather service over the internet and brings the answer back.

7. Assistant replies — in voice or text:

- ✓ Amazon Polly converts the response into natural-sounding speech (for voice interactions), while conversation itself — what was said and what was understood — is logged into a Logs & Context DynamoDB table.
- ✓ The assistant can remember context if the customer asks a follow-up question.

8. System keeps learning:

- ✓ Separately, and not tied to any single customer's request, a Train Model Lambda periodically reviews the logged conversations, retrains the underlying ML model, saves the improved version back into S3 "Brain" bucket.
- ✓ AWS Systems Manager helps coordinate when this retraining happens.

9. Data flows into Salesforce:

- ✓ Whenever there's something Salesforce needs to know - a new order, a feedback, a customer detail ,the lex-sfdc-dip Lambda function securely sends it over an encrypted connection (SSL/TLS) to Salesforce's API Factory.
- ✓ which then stores it in Salesforce's own Data Repository. From there, the company's sales and support teams can see and act on it inside Salesforce itself, without ever touching AWS directly.